32-bit microcontrollers

FLASH erase and write protection function

Application Note

Rev1.01 June 2025

Applicable objects

Product series	Product model	Product series	Product model
L Series	HC32L110 HC32L130 HC32L136 HC32L072 HC32L073 HC32L170 HC32L176 HC32L190 HC32L196 HC32L166 HC32L186	**F S**eries	HC32F003 HC32F005 HC32F030 HC32F072 HC32F170 HC32F176 HC32F190 HC32F196 HC32F002 HC32F052 HC32F420
A Series	HC32A052	**-**	-

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Contents

1 Overview

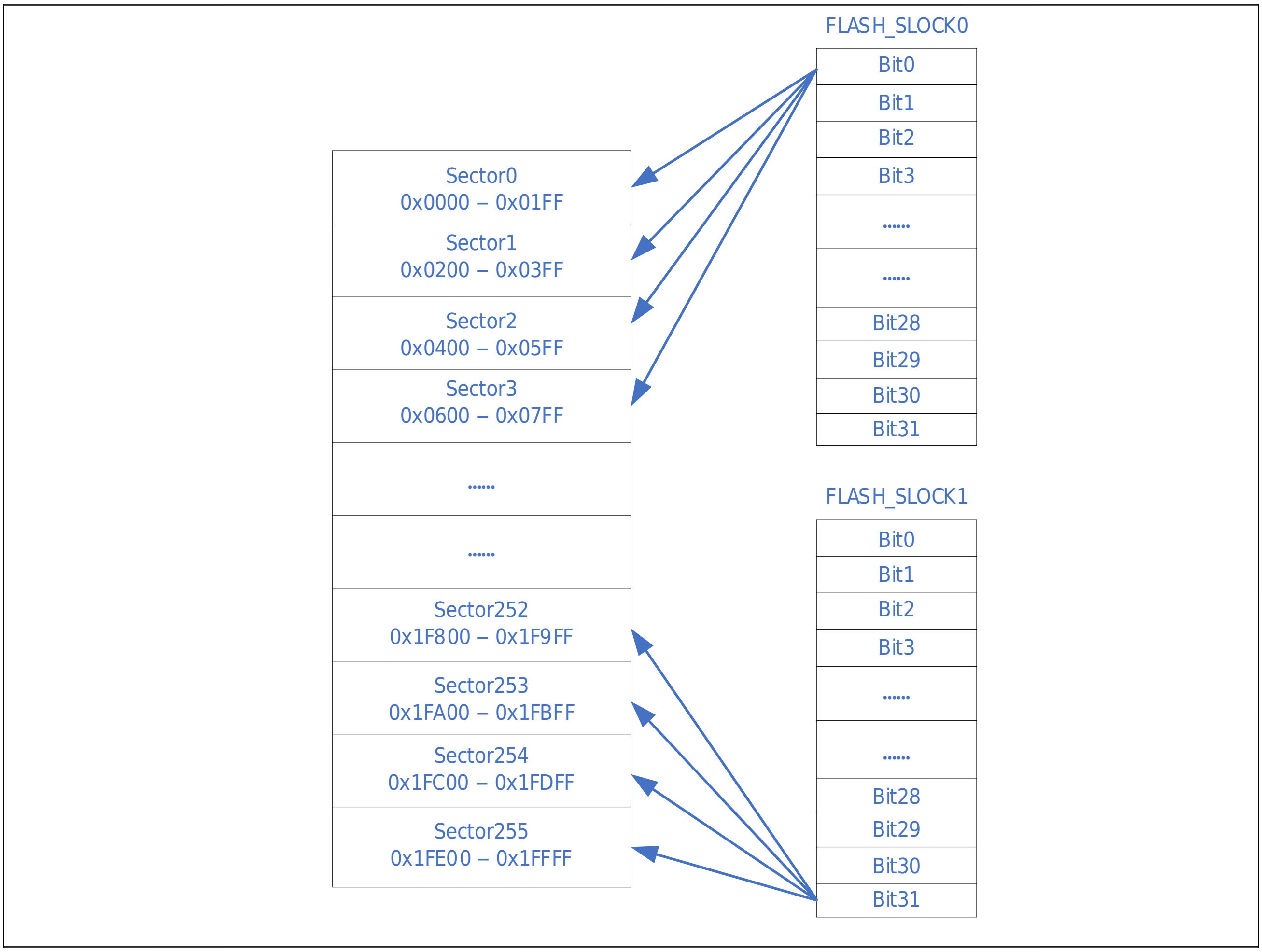
This document mainly introduces some precautions for using FLASH erase and write in the program of Xiaohua Semiconductor's specific series MCU.

2 FLASH erase protection

2.1 FLASH erase protection register

The FLASH module of the MCU series described in this article has an erase protection register. When the control bit of the register is 0, the corresponding page of the FLASH can be erased and written. The size of each FLASH page is 512 bytes. The whole chip is divided into several pages. Every 4 pages share an erase protection control bit of the erase protection register. Each series of MCU has 1~4 erase protection registers SLOCKx according to the size of FLASH.

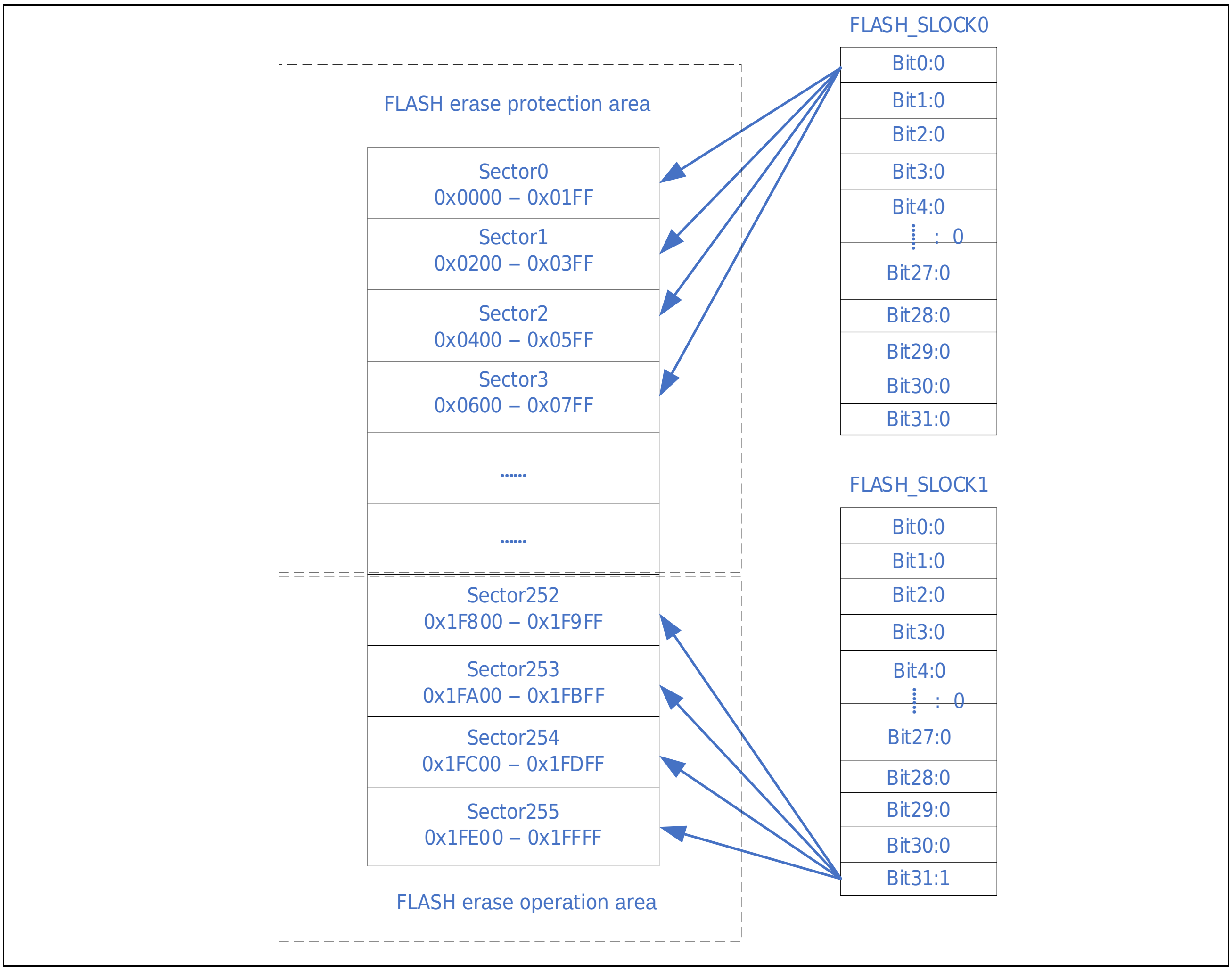
For example, the FLASH size in the figure below is 128KB, the sectors it has are 0~255, and the erase protection registers it has are FLASH_SLOCK0 and FLASH_SLOCK1.



2.2 Using erase protection register

When the program needs to erase or write certain pages of FLASH, the control bit of the erase protection register corresponding to the page needs to be set to 1 first: erase and write are allowed.

When releasing the erase protection and allowing erase and write (unlocking), it is strongly recommended that users only release the control bit corresponding to the page that needs to be erased and written, and the control bit of the erase protection register SLOCKx corresponding to the page in the user code area should never be set to 1.



When rewriting the erase protection register, it is recommended to check the correctness of the parameters written to the register. If the passed parameters are incorrect, the operation will not be performed and an error flag will be returned.

It is also recommended to check the correctness of the operation address during erase and write operations. If the address is not in the correct range, the operation will not be performed and an error flag will be returned.

The operation of the SLOCKx register also requires the BYPASS register to be operated first. Therefore, when the user operates these registers, if the interrupt is not masked, the value of the corresponding register needs to be queried after the operation is completed to see if the write is successful. If the write is unsuccessful, it is necessary to try to repeat the operation. It is also recommended to add a limit on the number of repeated operations to avoid the program being stuck due to continuous failure of the operation.

In addition, the operations of the following registers also require the BYPASS register to be operated first, so after the operation is completed, it is also necessary to query the readback confirmation and retry operations as mentioned above:

Parameter register, CR, WAIT (exist in some series) and CACHE (exist in some series).

3 Summary

The above chapters briefly introduce the application method of the FLASH erase and write protection function of Xiaohua Semiconductor MCU*. In the actual application development process, if the user needs to further understand the use and operation of this module, the corresponding reference manual should prevail. The samples and driver libraries mentioned in this chapter can be used as further experiments and learning for users, and can also be used as references in actual development.

Revision history

Revision	Revision Date	Revision Content
Rev1.00	2025/01/20	Initial version released.
Rev1.01	2025/06/04	Applicable objects: HC32A052 series models added.